MSTS CONSIST GENERATOR

USER MANUAL & OPERATIONAL QUICK START GUIDE

Official Reference
MSTS & Open Rails

WORKFLOW & STEP-BY-STEP INSTRUCTIONS (PART I)

STEP 01 Launch Application

Start the **MSTS Consist Generator** utility on your Windows PC via the executable or python launcher.

STEP 02 Browse Trainset Library

Under the **TRAINSET LIBRARY** section, click the "**Browse & Scan...**" button.

STEP 03 Select TRAINSET Directory

Navigate to your MSTS Trainset root folder, normally located at:
`C:\Program Files (x86)\Microsoft Games\Train Simulator\TRAINS\TRAINSET`
Important: Choose the `TRAINSET` directory itself. Do not select `TRAINS`, `TRAIN`, or an individual vehicle subfolder.

STEP 04 Verify Scan Results

Wait for the automated scanner. It will report total detected `.eng` (locomotive) and `.wag` (carriage/wagon) files. The scan reads immediate stock subdirectories; arbitrarily nested folders are bypassed.

STEP 05 Configure Consist Settings

Define production parameters inside the **CONSIST SETTINGS** panel:

- **Consists to create:** Total quantity of distinct consist (`.con`) files to assemble.
- **Base filename / name:** Prefix for files and internal names (avoid Windows-restricted characters like `\ / : * ? < > |`).
- **Locomotives per consist:** Maximum target count of `.eng` power units per consist.
- **Passenger / coach count (.wag):** Number of carriages assigned to passenger service groupings.
- **Freight / extra count (.wag):** Additional rolling stock units added from the shared asset catalog.
- **Maximum vehicles:** Absolute ceiling cap on total connected units for each consist.
- **Random seed (optional):** Leave blank for true randomness, or enter an integer seed for repeatable formations.
- **MaxVelocity values:** Maintain established defaults unless specialized physics profiles are required.

STEP 06 Choose Stock Mix Mode

- **Mixed:** Draws randomly across the complete wagon catalog with power dictated by Locomotive Policy.
- **Passenger / Freight:** Heuristic filters based on common naming patterns (inspect output formations manually).
- **Rolling stock only:** Tailored for unpowered rakes/sidings; requires selecting *No locomotives* in step 8.

STEP 07 Toggle Optional Assembly Flags

- **Randomize counts:** Dynamically varies locomotive and wagon totals between 0 and your configured limits.
- **Avoid repeats where possible:** Enforces distinct rolling stock choices per train until unique assets are exhausted.
- **Random wagon Flip ( ):** Introduces the MSTS reversal token on selected wagons for realistic visual variation.

Pro Tip: For large installations containing thousands of assets, set a **Random Seed** to document and reproduce favorite activity formations accurately across multiple test runs.

WORKFLOW & STEP-BY-STEP INSTRUCTIONS (PART II)

STEP 08 **Set Formation Logic & Locomotive Policy**

Select locomotive allocation: **Every consist** (full allocation), **Some consists (50%)** (power added to ~half), **Random (0 to maximum)**, or **No locomotives** (creates pure rolling stock sets).

STEP 09 **Configure End-of-Train Tail Vehicles**

Enable "**Append SLR / BVZI / caboose / guard van at end**" to attach brake vans. Choose frequency: *Every eligible consist*, *Some consists (50%)*, or *Random*. Keyword scanning detects `SLR`, `BVZI`, `caboose`, `guard van`, or `brake van`.

STEP 10 **Generate Consists**

Click "**GENERATE CONSISTS**". Review the execution summary dialogue and inspect the newly written consist file catalog.

STEP 11 **Open and Inspect Output Folder**

Click "**Open Output Folder**". Files are exported to your TRAINSET parent directory: `<TRAINS>\MSTS_Generated_Consists`. Copy valid `.con` files into your simulator's active `CONSISTS` folder.

STEP 12 **In-Game Validation**

Always test generated files in MSTS or Open Rails explore mode before embedding them in complex activities or routes. Verify textures, shapes, and coupling direction.

SUGGESTED FIRST TEST **Recommended Verification Preset**

Scan TRAINSET • Consists to create = **2** • Base filename = **Test_Consist** • Locomotives per consist = **1** • Passenger count = **4** • Freight count = **0** • Maximum vehicles = **20** • Stock mix = **Mixed** • Locomotive policy = **Every consist** • Leave optional checkboxes *unchecked*. Generate, inspect in the output folder, and test in simulator.

TROUBLESHOOTING GUIDE

"No .eng/.wag files found"

Ensure you selected the root `TRAINSET` directory, not parent folders. Scanner checks top-level subfolders only.

Output folder cannot be created

Run utility with administrator rights or move simulation assets out of protected UAC Windows directories.

Generated consist does not load

Inspect `.con` references in a text editor to confirm target `.s`, `.sms`, and cab assets exist on disk.

Tail vehicle (SLR/Caboose) missing

Ensure your tail-stock files or directory names include standard markers: `SLR`, `BVZI`, `caboose`, or `brake van`.

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